

14

The population of different trees in a field increased by 10 % in first year, increased by 8 % in second year and decreased by 10 % in third year. If at present the number of trees is 26730, then find the number of trees in the beginning.

- a. 30000 b. ~~25000~~ c. 27000 d. 27865

$$P \times \frac{110}{100} \times \frac{108}{100} \times \frac{90}{100} = 26730$$

Handwritten annotations above the equation:

- 10 above the first fraction
- +2 above the second fraction
- 4 above the third fraction
- 27 above the first fraction
- 297 above the second fraction
- 3 above the third fraction

Handwritten annotations below the equation:

- 25 below the first fraction
- 8 below the second fraction



14

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13 *Ans*

The present population of a country estimated to be 10 crores is expected to increase to 13.31 crores during the next 3 years the uniform rate of growth is?

$$\underline{\underline{10\%}}$$

$$10 \times (\underline{\underline{M.F}})^3 = 13.31$$

$$(M.F) = \sqrt[3]{\frac{1331}{1000}} = \frac{11}{10} \curvearrowright$$

$$\frac{11}{10} \times 100 = 110\% \uparrow$$



12. The value of lathe machine depreciates at the rate of 10 % per annum. If the cost of machine at present is Rs. 160,000, then what will be its worth after 2 years?

a. Rs. 122,365 b. Rs. 153,680 c. ~~Rs. 129,600~~ d. Rs. 119,900

$$1,60,000 \times \frac{90}{100} \times \frac{90}{100} = ?$$

$$\underline{16 \times 81} = 1296$$



✓
11Q. The population of a town is 18000. It increases by 10% during first year and by 20% during the second year. The population after 2 years will be ?

(A) 19800

(B) 21600

(C) 23760

(D) None

$$18000 \times \frac{110}{100} \times \frac{120}{100} = \underline{\hspace{2cm}}$$



10

Q. Journalist Popatlal's income is 37.5% less than Dr. Iyyer's. Then Dr. Iyyer's income is how much % more than Popatlal's income?

$$- \frac{3}{8} //$$

$$- \frac{P}{5}$$

$$\frac{+3}{5} \times \frac{20}{100} = 60\%$$



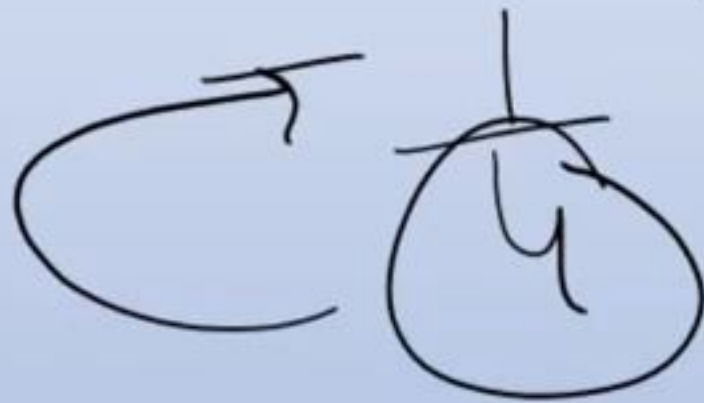
9Q. If A's salary is 25% less than B's salary, then B's salary is what % more than A's salary?

1) Cannot be determined
5) 20

2) 21

~~3) 33.33~~

4) 25



A ↓
3

B ↓
4

$$\frac{+1}{3} \times 100$$



Q If Arjun's salary is 20% more than that of Bheem, then how much percent is Bheem's salary less than that of Arjun?

- 1) ~~16.66%~~ 2) 20% 3) 40% 4) 10% 5) None of these

$$\left[\frac{20}{120} \times 100 \right]$$

$$\begin{array}{c} \text{Arjun} \\ \hline \downarrow \\ 120 \end{array}$$

$$\begin{array}{c} \text{Bheem} \\ \hline \downarrow \\ 100 \end{array}$$

$$\frac{-20}{+20} \times 100$$



8Q If Arjun's salary is 20% more than that of Bheem, then how much percent is Bheem's salary less than that of Arjun?

- 1) ~~16.66%~~ 2) 20% 3) 40% 4) 10% 5) None of these

$$\left[\frac{R}{100 \pm R} \times 100 \right]$$

Arjun
↓

120

Bheem
↓

100

$$\frac{-20}{+20} \times 100$$

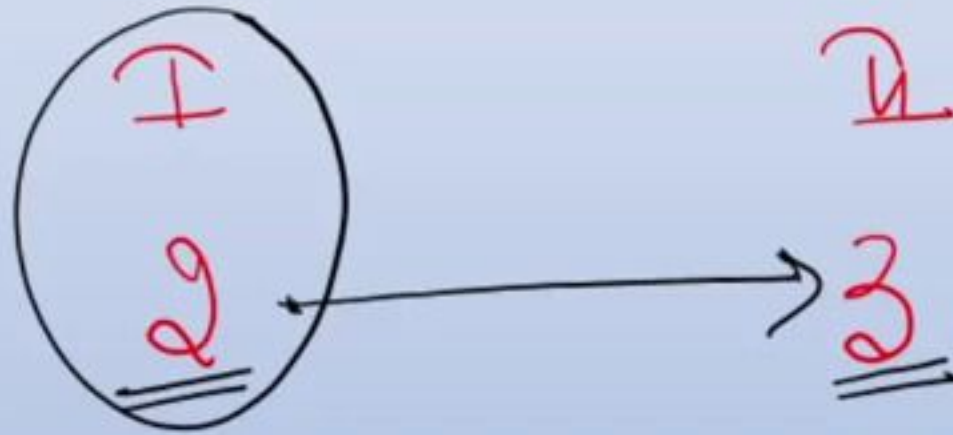


7

1. What percent is 2 minutes 24 seconds of an hour?

a. 6%
c. 4%

b. 2%
d. 8%



$$\% \text{ change} = \frac{+1}{2} \times 100 = 50\%$$



CONCEPT 02:

1. $\text{Percentage } \underline{\text{Change}} = \frac{\text{Final} - \text{Initial}}{\text{Initial}} \times 100$

2. Quantity I is how much percent of Quantity II

$$\frac{\text{Quantity I}}{\text{Quantity II}} \times 100$$

3. Quantity I is how much percent more than Quantity II

$$\frac{\text{Quantity I} - \text{Quantity II}}{\text{Quantity II}} \times 100$$

4. Quantity I is how much percent less than Quantity II

$$\frac{\text{Quantity II} - \text{Quantity I}}{\text{Quantity II}} \times 100$$





$\uparrow \left\{ \begin{array}{l} \underline{20\% \uparrow} \\ \underline{30\% \uparrow} \end{array} \right.$
 $\downarrow \left\{ \begin{array}{l} \underline{40\% \downarrow} \\ \underline{20\% \downarrow} \end{array} \right.$

$$\begin{array}{r}
 \times \frac{120}{100} \\
 \hline
 \times \frac{130}{100} \\
 \hline
 \times \frac{90}{100} \\
 \hline
 \times \frac{80}{100}
 \end{array}$$



$$20\% \uparrow = \frac{1}{5} \quad \left(\times \frac{6}{5} \right)$$

$$20\% \downarrow = \frac{1}{5} \quad \left(\times \frac{4}{5} \right)$$

$$\begin{array}{l}
 33.33\% \uparrow \downarrow = \frac{1}{3} \quad \left(\times \frac{4}{3} \right) \\
 \left(\times \frac{2}{3} \right)
 \end{array}$$





- A and B appeared in an examination . If the difference of their marks is 25 and percentage of their marks is given as 20 % . Find the full marks for which examination has been held ?

$$\frac{25}{20} \times 100$$

- If the price of one Kg of rice is increased by 25%, the increase is Rs 12. Find the new Price of rice per kg ?

$$25\% = \frac{1}{4}$$

$$\begin{array}{l} I_1 = 4 \\ I_2 = 5 \end{array} \quad \left. \begin{array}{l} \\ \end{array} \right\} 1 = 12$$

$$\underline{\underline{60 \quad 25}}$$



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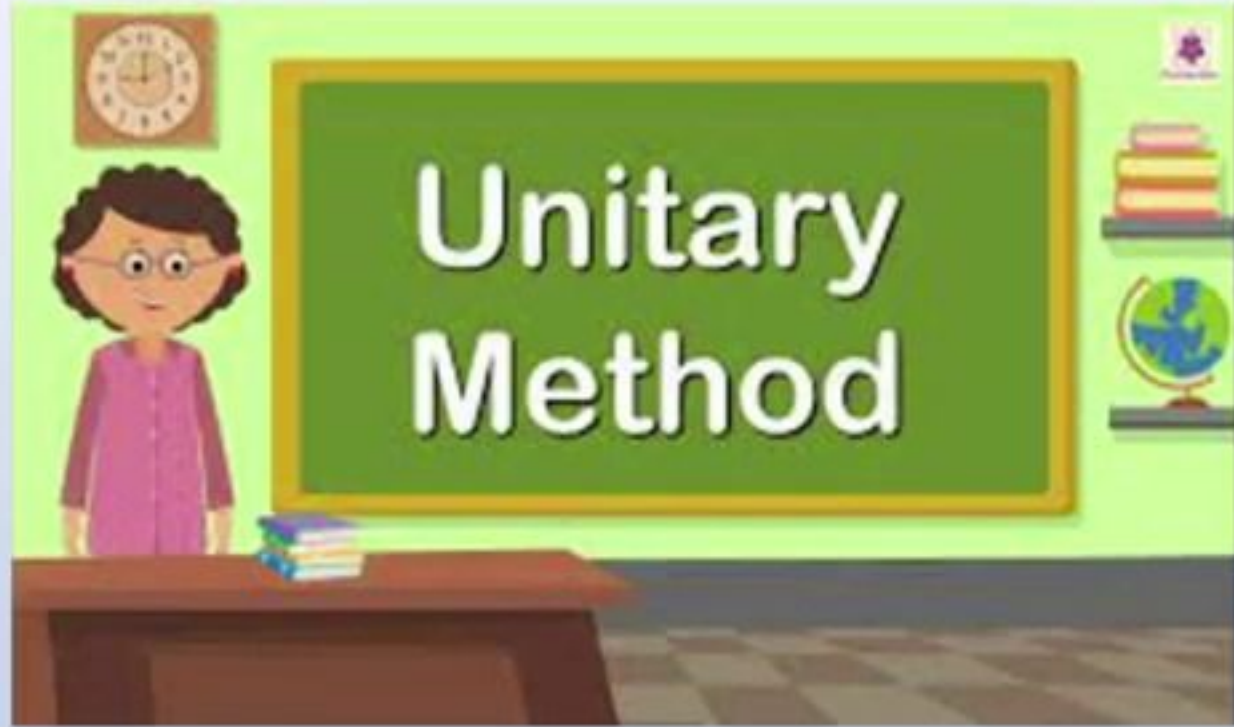


- If the price of one Kg of wheat is increased by $12\frac{1}{2}\%$, the increase is Rs.5 . Find the original and new price of wheat per Kg ?

$$12\frac{1}{2}\% = + \frac{1}{8}$$

$$\begin{array}{l} I_1 = 8 \\ I_2 = 9 \end{array} \quad \left. \begin{array}{l} \xrightarrow{40\%} \\ \xrightarrow{45\%} \end{array} \right\} I = 5$$





25% of a number is 20. what is 40% of that number also find the number ?

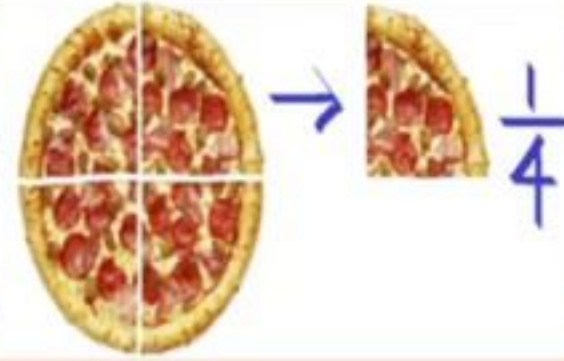
$$\frac{20}{25} \times 100 = 80$$



CONCEPT 01

1205
 $\frac{1}{4} \times 2 = \frac{1}{2}$

$\frac{1}{2} = 50\%$	$\frac{1}{8} = 12.50\%$ or $12\frac{1}{2}\%$
$\frac{1}{3} = 33.33\%$ or $33\frac{1}{3}\%$	$\frac{1}{9} = 11.11\%$ or $11\frac{1}{9}\%$
$\frac{1}{4} = 25\%$	$\frac{1}{11} = 9.09\%$ or $9\frac{1}{11}\%$
$\frac{1}{5} = 20\%$	$\frac{1}{12} = 8.33\%$ or $8\frac{1}{3}\%$
$\frac{1}{6} = 16\frac{2}{3}\%$	$\frac{1}{14} = 7.14\%$ or $7\frac{1}{7}\%$
$\frac{1}{7} = 14.28\%$ or $14\frac{2}{7}\%$	$\frac{1}{15} = 6.66\%$ or $6\frac{2}{3}\%$



NOTE :

$$\frac{1}{3} = 33\frac{1}{3}\% \Rightarrow \frac{2}{3} = 66\frac{2}{3}\%$$

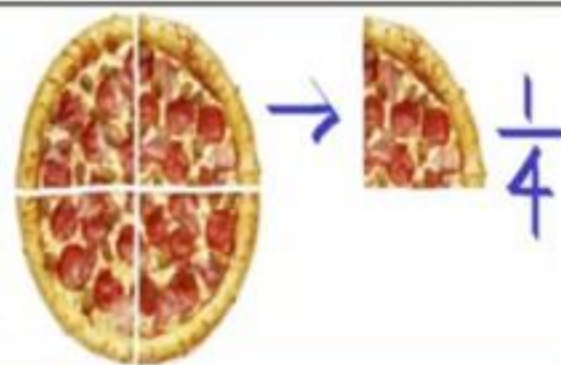
$$\frac{1}{8} = 12.5\% \Rightarrow \frac{3}{8} = 37.5\%$$

$\frac{1}{8} = 12.5$ $\frac{1}{12} = 8.33\%$



CONCEPT 01

$\frac{1}{2} = 50\%$	$\frac{1}{8} = 12.50\%$ or $12\frac{1}{2}\%$
$\frac{1}{3} = 33.33\%$ or $33\frac{1}{3}\%$	$\frac{1}{9} = 11.11\%$ or $11\frac{1}{9}\%$ ✓
$\frac{1}{4} = 25\%$	$\frac{1}{11} = 9.09\%$ or $9\frac{1}{11}\%$ ✓
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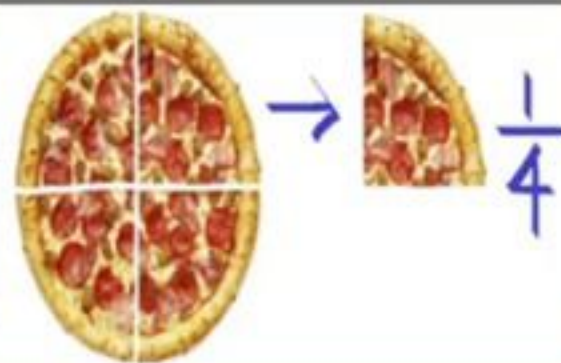
$$\frac{1}{9} = 11.11\%$$

$$\frac{1}{11} = 09.09\%$$



CONCEPT 01

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